

Topic 2. AUTOMOTIVE SYSTEMS.

If GM had kept up with technology like the computer industry has, we would all be driving \$25 cars that got 1,000 MPG (Bill Gates)



The air coming out of the modern diesel engine is cleaner than air coming in it

(contemporary joke)

SPEAKING: Autonomous Driving

Please study the following links:

Elon Musk makes clear his stance on self-driving cars, AI oversight, and his 'ad for Mars' (a businessinsider.com article, 2020)

<https://www.businessinsider.com/elon-musk-on-self-driving-cars-artificial-intelligence-and-mars-2020-12?r=DE&IR=T>

Ten ways autonomous driving could redefine the automotive world (2015 article about 2000-2050 trends in AD)

<http://www.mckinsey.com/industries/automotive-and-assembly/our-insights/ten-ways-autonomous-driving-could-redefine-the-automotive-world>

4 Major PROBLEMS for SELF-DRIVING CARS! (2021, 9 min 25 sec))

https://www.youtube.com/watch?v=kPcP2_xPHps

Beyond Tesla: Driverless Startups Promise Next-Level Autonomous Vehicles | WSJ (2021, 8 min)

<https://www.youtube.com/watch?v=UdOxt11ofjQ&t=35s>

<https://www.synopsys.com/automotive/autonomous-driving-levels.html>

<https://pca.st/4zfr96u0> (Episode: Pork Pie for Your Ears: ...Tesla Autopilot Casualties, June 12, 2023. This Week in Tech podcast). It is also available as a video

<https://youtu.be/88r9CFI501s?si=AZjHg5PZRIYDnz3c>

Listen to 1:14:00-1:50:30. It is a good example of a discussion even though it is quite informal. Your discussions should be more structured and formal.

Your teacher will provide you with a setting.

Take time to explore the following key terms — understanding and using them will be essential for the upcoming discussion.

Autonomous Driving Vocabulary for Discussion

Term	Definition
Actuator control	The direct control of steering, throttle, braking, and gear shifting through electronic control units based on navigation and perception input.
ADAS	Advanced Driver Assistance Systems; semi-autonomous features like lane keeping, adaptive cruise control, and emergency braking.
ASIL (Automotive Safety Integrity Level)	A risk classification system under ISO 26262 used to define safety requirements based on hazard severity and exposure (A, B, C, D, with D being the most critical).
Autonomy Levels (SAE Levels of Driving Automation)	Autonomy levels refer to the six-tier classification system (Levels 0–5) defined by the Society of Automotive Engineers (SAE International) in the SAE J3016 standard. This framework specifies the degree of driver involvement and system capabilities in vehicles with driving automation features.
Camera-based perception	The use of visual data from optical sensors to detect lane markings, signs, objects, and pedestrians using computer vision algorithms.
Closed-loop testing	Testing where the output of the system feeds back into the input in real time, mimicking real-world decision cycles.
Dead reckoning	Estimating current position based on a known starting point and motion data when GPS is unavailable.
Deep learning/neural networks	Machine learning models that mimic the human brain's structure and are trained to recognise patterns in complex sensory data.
Edge computing	The processing of data close to the source (i.e., in the vehicle) to reduce latency and enable real-time decision-making.
Embedded systems	Specialized computing systems within the vehicle that perform dedicated control functions under real-time operating constraints.
Ethical decision-making	Algorithmic decision processes that address moral dilemmas and prioritise human safety and fairness.
Fail-operational system	A system that continues to operate safely even after the failure of one or more of its components.
Functional safety	The discipline ensuring that a system remains safe under defined conditions and faults, often guided by ISO 26262 standards.
Geofencing	Virtual geographic boundaries that restrict where autonomous vehicles can operate.
HD map	High-definition map with centimetre-level precision containing lane geometry, road rules, and traffic infrastructure data.
HMI (Human-Machine Interface)	The interface through which humans interact with the vehicle's systems, including visual displays and tactile controls.

ISO 26262	An international standard for the functional safety of electrical and electronic systems in production automobiles.
Latency	The delay between sensing an event and responding to it, crucial for collision avoidance and control accuracy.
LiDAR	Light Detection and Ranging; a remote sensing method using pulsed laser light to measure distances and create high-resolution 3D maps of the vehicle's surroundings.
Localization	Determining the vehicle's precise position within a map using GPS, inertial sensors, and SLAM algorithms.
Motion planning	Generating a set of feasible trajectories that account for physical constraints, traffic laws, and real-time data to guide the vehicle's motion.
Object detection	A real-time process in which AI models identify and classify objects such as cars, cyclists, and pedestrians in the driving environment.
Operational Design Domain (ODD)	The specific conditions under which an autonomous system is designed to function, including weather, geography, and traffic types.
OTA updates	Over-the-air software updates pushed to the vehicle to fix bugs, add features, or enhance safety without a workshop visit.
Path planning	The algorithmic process of determining a safe and efficient route from the vehicle's current location to its destination, avoiding dynamic and static obstacles.
Perception stack	The set of software layers responsible for interpreting sensor data and understanding the environment.
Radar	Radio Detection and Ranging; uses radio waves to detect the position, velocity, and trajectory of surrounding objects, even in poor visibility conditions.
Real-time operating system (RTOS)	An OS that guarantees timing constraints are met for processing tasks critical to vehicle control.
Redundancy architecture	The inclusion of backup systems and fail-safe mechanisms to ensure continued operation in case of a component failure.
Safety validation	A formal process to prove that an autonomous system meets required safety performance standards in all expected scenarios.
Semantic segmentation	A computer vision task that assigns a class label to each pixel in an image, enabling detailed scene understanding.
Sensor fusion	The integration of data from multiple sensors (LiDAR, radar, cameras) to create a comprehensive and accurate model of the environment.
Simulation environment	A virtual testing ground that replicates real-world conditions for training and validating autonomous systems.
SLAM	Simultaneous Localization and Mapping; an algorithmic technique for constructing a map of an unknown environment while tracking the agent's position within it.
Teleoperation	Remote control of autonomous vehicles by a human operator, typically in challenging or unstructured environments.
Throughput	The amount of data processed by the system over time; critical for high-volume sensor input in AVs.

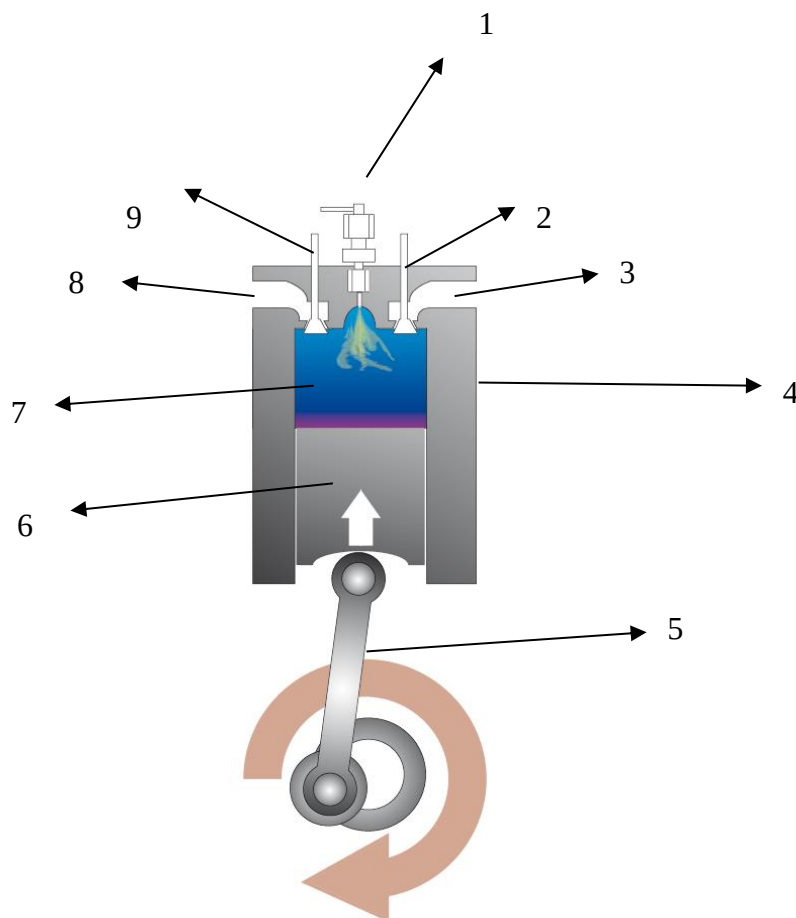
Trajectory optimization	A computational method to refine motion plans for minimal energy use, maximum safety, and passenger comfort, considering kinematic constraints.
Trolley problem	A hypothetical ethical scenario involving a forced choice between two harmful outcomes, used in discussions of AV ethics.
V2I communication	Data exchange between vehicles and road infrastructure, such as traffic signals and smart signage.
V2V communication	Exchange of information between vehicles to coordinate behaviour and prevent collisions.
V2X communication	Vehicle-to-everything communication, including V2V (Vehicle-to-Vehicle), V2I (Vehicle-to-Infrastructure), and V2N (Vehicle-to-Network).

WATCH THE FOLLOWING VIDEO

The Differences between Petrol and Diesel Engines (4 min 38 sec)

<https://www.youtube.com/watch?v=rlK7JIAz9WY>

Exercise 1. Label the illustration with the terms you find below. ¹



combustion chamber

injector

exhaust valve

piston

¹ Source: Adapted from http://www.planet-schule.de/fileadmin/dam_media/wdr/meilensteine_natur_technik/diesel/m3_diesel_funktion_3.0.pdf

[illegible]

LANGUAGE DEVICES: Contrasting

1. These devices start a new sentence and refer back to the previous one: **however, nevertheless, on the other hand:**

Gas and diesel engines both convert fuel into energy through a series of small explosions or combustions. **However**, these engines have their differences.

When gasoline engines hit the 120,000-150,000 mile mark, the cylinders will start to show some wear and the efficiency will decrease. Diesel engines, **on the other hand**, are known for their long life cycle.

Autopilot systems are useful. **Nevertheless**, some specialists consider them to be dangerous.

2. These devices contrast two clauses within one sentence: **while, despite the fact that, in spite of the fact that, even though, whereas:**

In a gas engine a spark ignites the fuel and air mixture, **whereas** a diesel engine does not use a spark.

While diesel engines are designed similarly to gas engines, the cylinder on a diesel engine has a removable liner.

Even though diesel fuel often costs more than gasoline, it contains more potential energy.

General business idioms that have to do with driving and traffic.

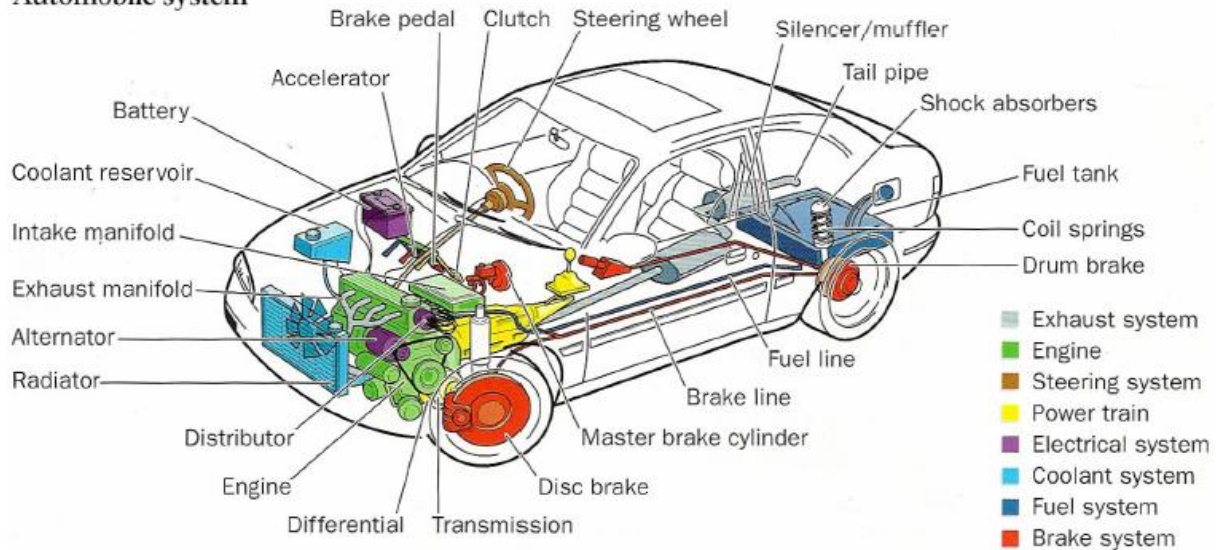
Use the following words to complete the sentences with the correct idioms:

to take a *back seat* (to somebody) ⚡ to shift into a higher gear ⚡
to give (somebody) the green light ⚡ to hit the road ⚡ to get a lot of mileage (out of something) ⚡

1. We are going to have to _____ if we want to finish the project by March.
2. Leslie is in charge of this project. Rick will have to _____ to her.
3. I am going to have to _____. My client is expecting me in half an hour.
4. I am waiting for the board of directors to _____ for the project.
5. He is _____ the fact that he is a boss's nephew.

TOPIC RELATED VOCABULARY

Automobile system

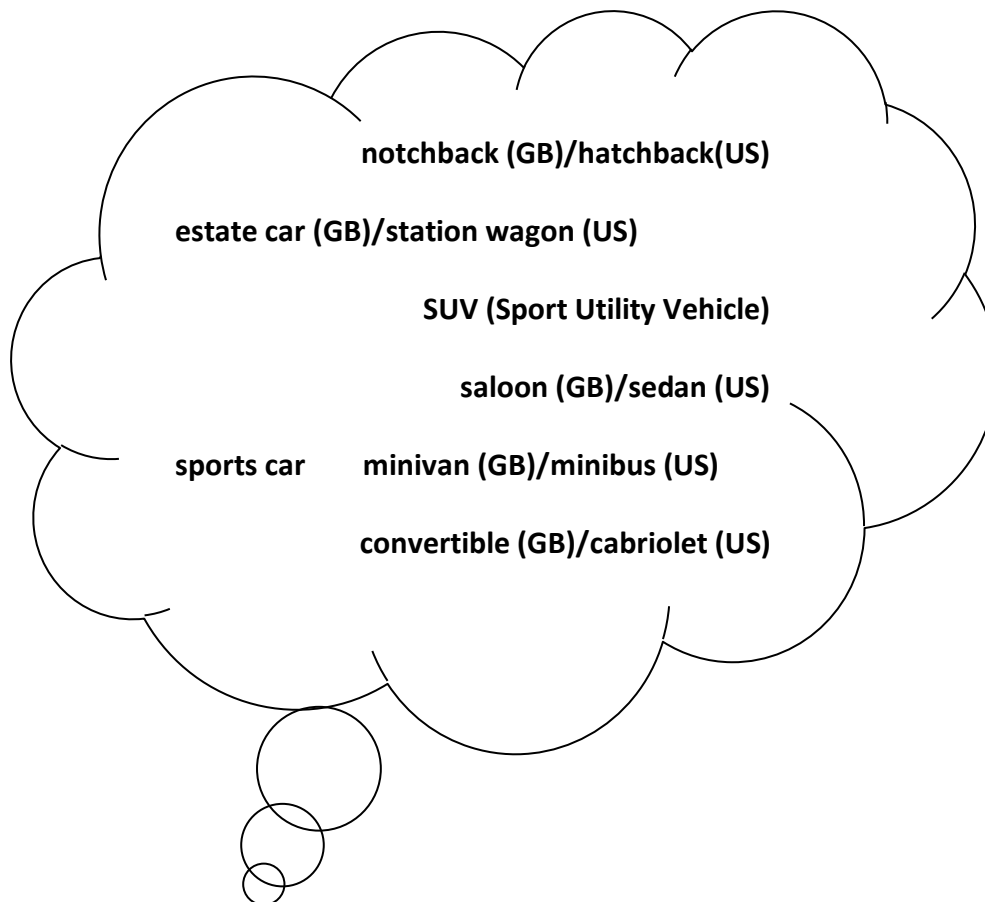
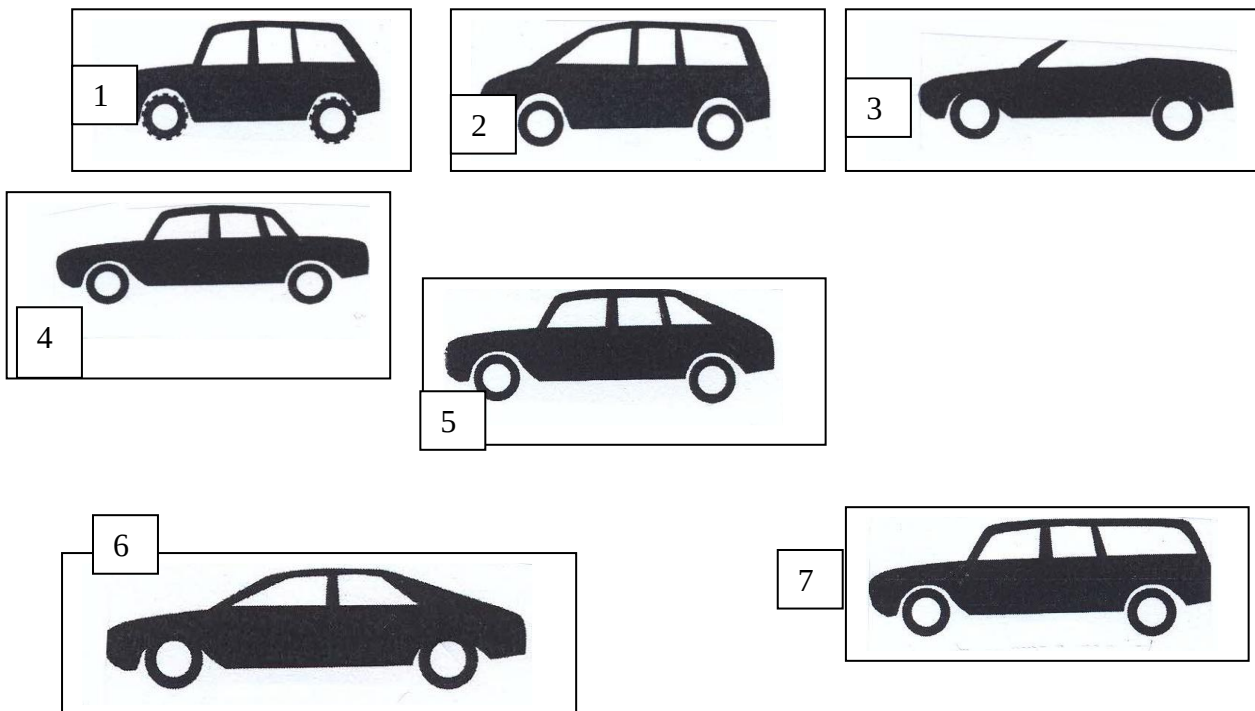


Match the part of the car with its function.

steering wheel	holds brake fluid
exhaust manifold	provides the power
radiator	stores electricity
fuel tank	ensures that the driven wheels turn at a different speed to each other when a car corners
brake line	produces electricity
silencer/muffler	sends an electric current to the spark plugs
battery	carries waste gases to the exhaust pipe
clutch	makes the car go faster when it is pressed
differential	used by the driver to turn the car
engine	holds fuel
brake cylinder	cools water from the engine
accelerator	connects the brake cylinder to the brakes
distributor	reduces the exhaust noise
alternator	disconnects the engine from the gearbox while the gears are changed

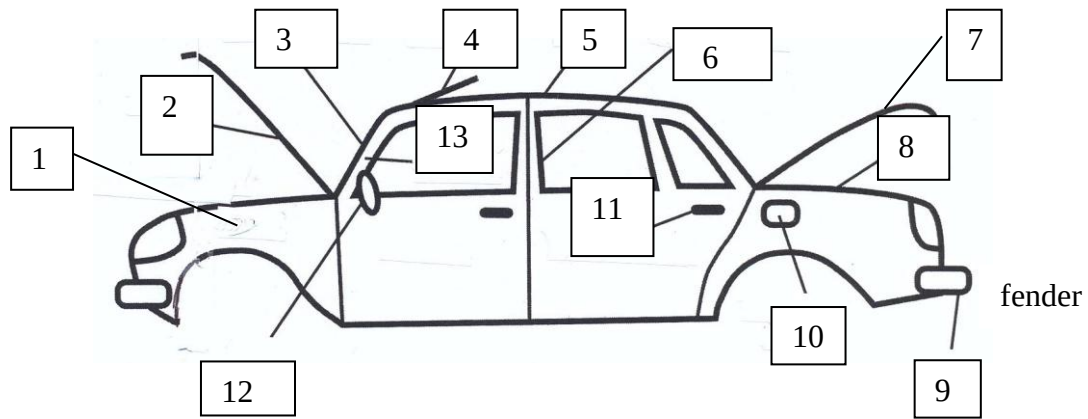
Car Body(ies)²

Exercise 1



² Source reference: Adapted from *engine – Englisch für Ingenieure* (3/2008, p.25) Pictorial / Source reference: Adapted from *engine – Englisch für Ingenieure* (1/2009, p.21) Pictorial (including pictorials on page 39-41)

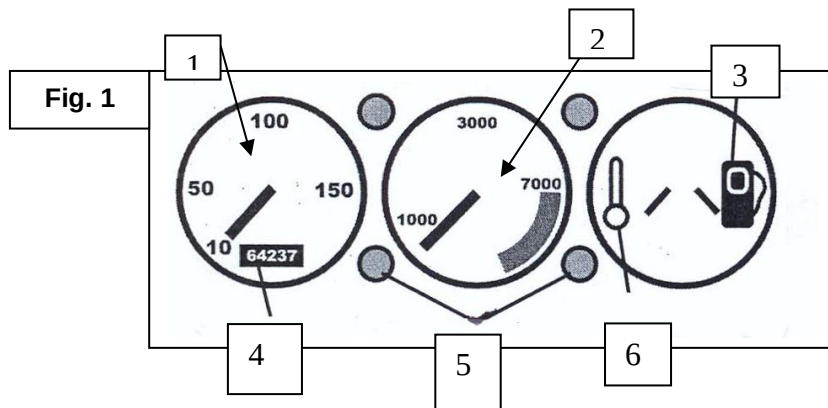
Exercise 2. Label the body of the car in the illustration below. **Use** the terms in the box. The first one has been done for you.

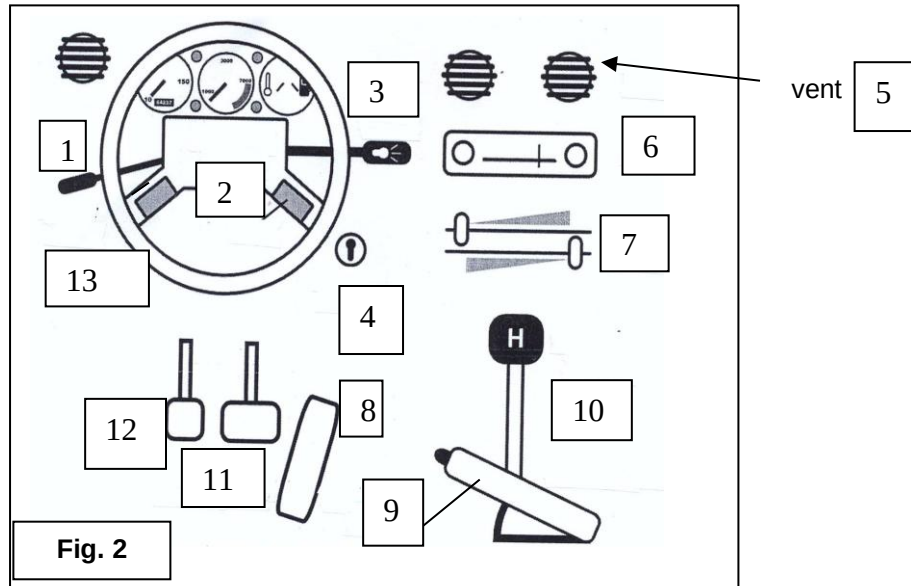


fuel tank flap...bonnet (AmE hood)...A-pillar...sunroof...
outside rearview mirror... wing (AmE fender)... door
handle...roof...boot (AmE trunk)...door pillar/B-pillar...
boot lid (AmE trunk lid)... fender (AmE bumper)...windscreen
(AmE windshield)

Exercise 3. Specify the parts in Fig. 1 and 2 with the help of the words in the box.

mileometer...warning light(s)...temperature gauge...
fuel gauge...speedometer...tachometer



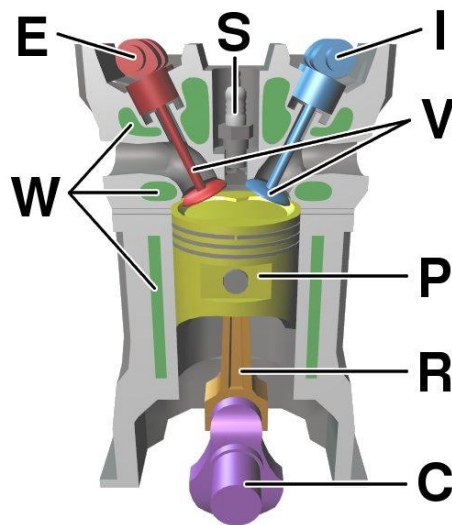
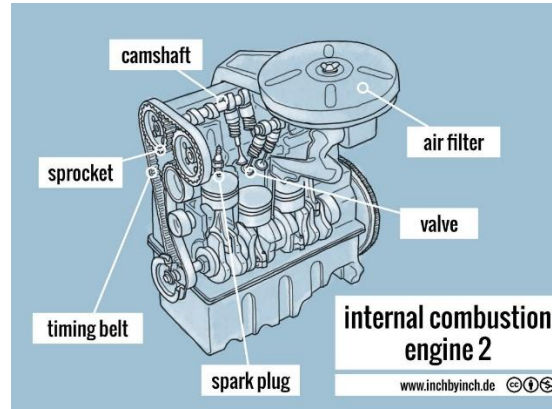
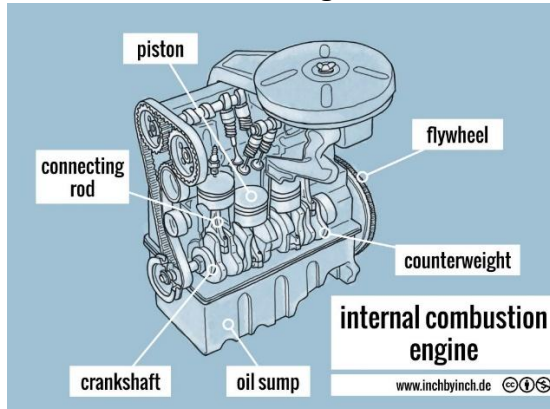


indicator (stalk)...steering wheel...light switch...radio...climate control...gear
 lever...handbrake (lever)...accelerator pedal...brake pedal...clutch
 pedal...ignition...horn

READING

TEXT 1.

Internal Combustion Engines



E = exhaust camshaft
I = inlet camshaft
V = valves (red=exhaust, blue=intake)
P = piston
R = rod
S = spark plug
C = crankshaft
W = cooling water jacket

Source: By User:Wapcaplet - Own work, made with Blender, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=182044>
Retrieved: 2021-12-13

In an internal combustion engine thermal energy is converted into mechanical energy. The thermal energy is released when the fuel is burned.

The piston completes four different strokes while turning the crankshaft.

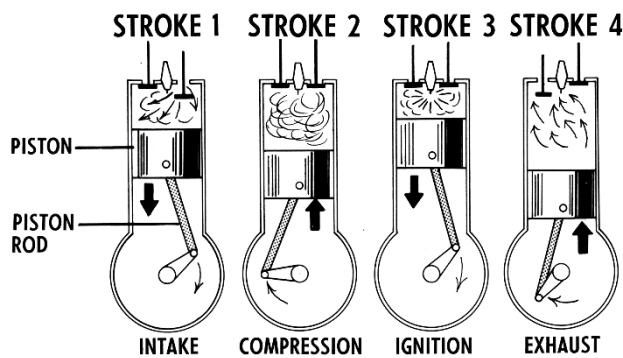
The four strokes in a petrol engine are:

Intake stroke: while the intake valve is open, the descending piston draws fresh petrol-and-air mixture into the cylinder.

Compression stroke: while the valves are closed, the rising piston compresses the mixture.

Ignition stroke: the mixture is then ignited by the spark plug. The pressure of the gases of combustion forces the piston downwards

Exhaust stroke: the exhaust valve is open and the rising piston discharges the spent gases from the cylinder.



Source: [https://commons.wikimedia.org/wiki/Category:Four-stroke_cycle#/media/File:Stroke_\(PSF\).png](https://commons.wikimedia.org/wiki/Category:Four-stroke_cycle#/media/File:Stroke_(PSF).png) retrieved: 2021-12-13

A Diesel engine works in a similar way. Pure air is sucked into a cylinder, a piston compresses the air and the air heats up. When the piston has compressed the air in the cylinder as much as possible, Diesel fuel is injected. At this point the air is so hot that the mixture ignites by itself. The pressure from the explosion forces the piston downwards and work is performed. The piston is brought back up through the momentum of the crankshaft and expels the used air from the cylinder. On its way down it sucks in fresh air again and the cycle starts anew.

Exercise 1. Match the strokes and their descriptions

- | | |
|-----------------------|--|
| 1. Exhaust stroke | a) Air-fuel mixture is drawn in |
| 2. Ignition stroke | b) Air-fuel mixture is compressed |
| 3. Intake stroke | c) Spark plug firing, explosion pushes piston down |
| 4. Compression stroke | d) Piston pushes out burned gases |

Exercise 2. Describe the difference between the petrol and the Diesel engine

Martina Weddig

TEXT 2.

DIGITAL VALVE TRAIN LAB REPORT

The *valve train* is the "*respiratory organ*" of *combustion engines*: it manages the *aspiration* of fresh air and the *discharge* of *exhaust gases*. Today, only mechanically driven *camshafts* are used in series production for this purpose, often equipped with an additional mechanism, some of which are quite complex.

Patrik Soltic and his team at Empa's Automotive *Powertrain* Technologies laboratory, together with hydraulics specialist Wolfgang Schneider, invented and developed an electro-hydraulic valve train that is significantly more flexible than today's series production technology. The *valves* are *actuated* hydraulically and controlled electrically via a *solenoid coil*. As soon as a control *current* flows, a specially designed hydraulic valve opens, allowing hydraulic fluid to open the gas exchange valve to the *desired extent* in milliseconds *counter* to a *spring*. When the current is switched off, the gas exchange valve is closed again by the spring force and feeds a large part of the hydraulic energy required for opening back into the hydraulic system. The system *achieves* a significantly lower energy *requirement* over a wide operating *range* compared to camshaft-driven systems. Together with an optimized gas exchange, the fuel *consumption* of the test *spark-ignition* engine is about 20 percent lower than with conventional valve control using a *throttle* in combination with camshafts in the low load range typical for passenger cars.

By selecting the operating parameters, the opening and closing times as well as the valve lift for each cylinder can be freely chosen. This means that each engine operating condition can be varied from cycle to cycle, for example by intelligent load control, by selecting the *residual* gas quantity remaining in the cylinder, or by deactivating unneeded cylinders without the driver noticing. This makes the engine highly adaptable to new renewable fuels. In addition, alternative combustion concepts can also be implemented comparatively easily, for example, homogeneous *self-ignition*: a fuel-air mixture is *ignited* at the right moment without ignition sparks by setting the correct conditions towards the end of compression. The mixture is combusted almost without pollution.

The new valve train was put into operation in a passenger car engine powered by natural gas and *derived from* a VW 1.4l TSI engine. The valve train has been running on an Empa engine *test bench* since October 2018 and has already *survived* many millions of cycles in fired engine operation *flawlessly*.

Rainer Klose, www.empa.ch/web/s604/flexwork, retrieved on January 10, 2022.

Task 1: Access the article for the illustration and more information. What additional information have you found out?

Task 2: Provide the German translation to the *words in italics* from the text:

Valve -

respiratory organ –

combustion engine –

aspiration -

discharge -

exhaust gases –

camshaft –

powertrain –

to actuate –

solenoid coil –

current –

desired extent –

counter –

spring –

to achieve –

requirement –

consumption –

spark-ignition –

throttle –

residual –

to ignite –

self-ignition –
to derive from –
test bench –
to survive –
flawlessly -

Task 3: Read the definitions to some of the words in italics from the text. What are the words?

- A thick spiral of wire through which an electrical current passes –
- A rod in an engine that works to change circular motion into vertical or lateral motion –
- A device attached to a pipe or a tube which controls the flow of air or liquid through the pipe of tube -

Definition and class

Texts containing technical terminology frequently contain definitions and explanations.

1. Common words and expressions used in definitions or explanations:

- Printers **are** output devices.
- The term composting **refers to** a type of waste management.
- By active solar energy **we mean** the use of solar energy devices which heat water or buildings.

2. Some definitions and explanations give further distinguishing characteristics by means of defining relative clause.

- A computer is an electronic device **which/that processes information**.
- A forester is a person **who is trained in the care and use of forest areas**.
- The arithmetic-logical unit is the part of the CPU **where arithmetic and decision-making operations are done**.

Note: The relative pronouns used in this type of definition or explanation will be
who or **that** for people
when for a period of time
where for a place or location
which or **that** for things.

3. Another way of defining or explaining is to use a noun, a noun phrase, or a clause separated from the rest of the sentence by commas or dashes.

- Demand side energy management – **the use of conservation measures which counteract increases in demand for energy** - is essential if we want to save the planet.
- Turnkey systems, **complete hardware/software products which are ready for use**, are available from many suppliers.

4. A definition usually includes three parts: the term to be defined, the class it belongs to, and the characteristics which distinguish it from the other members of the group.

TERM	CLASS/CATEGORY	CHARACTERISTICS
Recycling is	a process	which reuses materials that would otherwise be wasted.
Hypnoanalysis is	a psychoanalytical technique	in which hypnosis is used to elicit information from a patient's unconscious mind.
Silicon is	a non-metallic element	with semiconductor characteristics.
A thermometer is	an instrument	for measuring temperature.
Open-source software is	computer software	whose programming code is freely available to users.
RDF is	an acronym	for refuse derived fuel, which refers to the use of combustible materials in garbage as a source of energy.

Common Terms for Class

device	An object cleverly or especially made for a special purpose, e.g., a thermostat is a device for maintaining a constant temperature. <i>A compass in a coil of wire is a simple device for detecting an electric current.</i>
apparatus	(pl. apparatus or apparatuses) Scientific equipment used in the laboratory for scientific work, usually consisting of various components. <i>The apparatus for distillation has assembled and is standing on the table.</i>
instrument	An object used by a person for making, doing, or effecting something, e.g., a musical instrument for making music; dissecting instruments for carrying out a dissection. The term scientific instrument implies that the device is used for detecting, measuring, or recording. Examples are Geiger counter, thermometer, barometer. Instruments used for qualitative measurements have the ending -scope , for quantitative measurements the ending -meter , and for recording quantitative measurement the ending -graph . E.g., a <i>spectroscope</i> examines a spectrum qualitatively; a <i>spectrometer</i> measures the wavelengths of light in a spectrum. A <i>barometer</i> measures the atmospheric pressure quantitatively; a <i>barograph</i> records the quantitative measurements of atmospheric pressure on a graph.
tool	An object which is held in the hand or hands, and is used to assist a person in doing manual work, e.g., a chisel, a screwdriver, and pliers. In general speech the difference between tool and instrument implies that the work carried out with an instrument is more delicate, and requires more knowledge. A machine tool is a power-operated tool, e.g., a lathe. A screwdriver is a hand tool . An implement is a tool which is not restricted to manufacture, cultivation (agriculture) or repair, e.g., <i>kitchen implements</i> .
mechanism	A structure of moveable parts which interact mechanically, e.g. the mechanism of a clock.
property	One of the ways in which an object, substance, material, force or wave motion can be recognized, by appearance or behavior. <i>Inertia is the <u>property</u> of matter that causes it to resist any change in its motion.</i>
type	<i>Coal is a <u>type</u> of fossil fuel which was formed from undecayed vegetable matter laid down in an ancient swamp.</i>
process	<i>Annealing is the <u>process</u> of heating and then cooling a material (e.g. metal) to remove strains in the material.</i>
technique	<i>Incineration is a <u>technique</u> which uses burning to dispose of refuse.</i>
unit	<i>The foot is a linear <u>unit</u> in the fps (foot-pound-second) system.</i>
acronym	<i>What does (the <u>acronym</u>) AQI stand for? It stands for <u>air quality index</u>, which is used to monitor the amount of ozone and other pollutants in the air.</i>
abbreviation	<i>The <u>abbreviation</u> "etc." stands for "et cetera". <u>Note</u>: Frequently acronyms are referred to as abbreviations. What does the abbreviation "e.g." mean? - It means "for example" (exempli gratia). What does the abbreviation "i.e." mean? - It means "id est" (German "d.h.").</i>

Exercise 1

Choose a class (left column) and find its corresponding definition (right column).

Class	Definition of Class
1. tool	A is a category of things, for instance, Diesel is a of fuel.
2. property	An is a short form of a word or phrase, for instance, Dr. for Doctor, etc. for etcetera, m for meter.
3. unit	A is a way in which an object, substance, material, physical phenomenon can be recognized, by behavior and appearance.
4. acronym	A is an object that is held in the hands or hand and is used to assist a person in doing manual work, for instance, a hammer, and a screwdriver.
5. abbreviation	A is a quantity chosen as a standard in terms of which other quantities may be expressed, for instance, meter.
6. machine	A uses mechanical power and has several parts, each with a definite function and together performing a particular task, for instance, a wind generator.
7. instrument	An is formed from the initial letters of other words and sometimes pronounced as a word, for instance, ASCII or NASA.
8. device	An is an object used by a person for making, doing, or affecting something, therefore, used for detecting, measuring, or recording.
9. type	A is an object made or adapted for a particular purpose, especially a piece of mechanical or electronic equipment, for instance, a smoke detector.

Exercise 2

To which class (see previous pages) do the following terms belong?

Calif.	natural gas
can opener	odor
color	ohm
cooking	oscilloscope
drill	PCB
drying	sanding
DSL	solar
GB	soldering
hygrometer	spanner
joule	VDU
level	water pump
micrometer	wiring
milling	yd

Exercise 3

Complete the sentences by finding the appropriate class (see previous page).

TERM	CLASS	CHARACTERISTIC
EPA is	an _____	for environmental protection agency.
An odometer is	an _____	for measuring the distance a vehicle has traveled.
A wrench is	a _____	used for tightening or loosening a nut.
Sand blasting is	the _____	of cleaning metal objects with sand or other abrasives projected from a nozzle at high speed.
The mho is	the _____	of electrical conductance now known as the Siemens.
Inertia is	the _____	of matter that causes it to resist any change in its motion.
Biological degradation is	a _____	in which organic compounds are decomposed into inorganic chemicals.
Laser is	an _____	for Light Amplification by Stimulated Emission of Radiation.
A still is	an _____	to distill alcohol.
A hydrometer is	an _____	to determine specific gravities of liquids and therefore the strength (e.g., saline solutions).
A funnel is	a hollow cone-shaped _____	with a tube at the narrow end for pouring materials (usually liquids) through a small opening.

NOTE: Definitions are given in the present simple tense and use the verb to be. The indefinite article "a" is usually used for the term, although the definite article "the" is used when referring to a specific term.

Exercise 4³

Look at the pictures with the basic tools. Use the words from the box to label the corresponding picture.

drill...chisel...spanner...pliers...Allen key...fretsaw



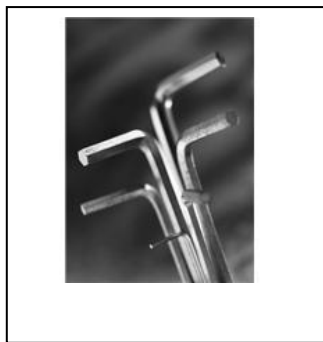
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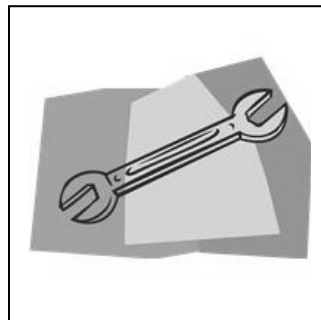
2



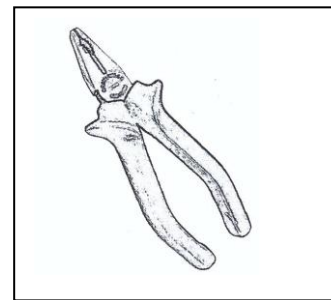
3



4



5



6

IDIOM: *to put/throw a spanner in the works*

to cause something to not go as planned

We just started dating when the corona virus restrictions **threw a spanner** in the works.

³ <http://office.microsoft.com/en-us/images/results.aspx?qu=chisel&ex=1#ai:MC900252275>, <http://office.microsoft.com/de-de/images/results.aspx?qu=bohrer&ex=1#ai:MC900382626>, <http://office.microsoft.com/de-de/images/results.aspx?qu=s%C3%A4ge&ex=1#ai:MC900340412>

Exercise 5

Read this example of a definition.

A thermometer is an instrument for measuring temperature.

Structure of a Definition:

.....

Now write definitions for three of the following terms.

goggles...windscreen...plastic...engine...hammer...drill

.....

.....

.....

Exercise 6

Complete the following sentences. You will need to use words already introduced and new words.

1. A compass is an **instrument** consisting of two arms joined at a pivot that is used to draw circles or arcs. A protractor is an instrument **used to** draw or measure angles on paper.
2. A seismograph is a(n) _____ used to record the strength of earthquakes.
3. Nuclear fusion is a _____ in which hydrogen nuclei are joined together to produce heat.
4. A film is a thin _____ of any substance, such as grease.
5. A rivet is a metal _____ with a head at one end which is used to fasten two pieces of metal or other material together. The rivet is passed through holes drilled in the pieces and the headless end is hammered flat.
6. A _____ is a protective device in electrical circuits or an apparatus. It consists of a thin piece of metal that melts if the current exceeds a particular value, thus breaking the circuit.
7. A cubic centimeter is a _____ of volume.
8. The _____ for *inch* is "in."
9. Clear cutting is a forest harvesting _____ in which all trees in an area are cut at one time.
10. A non-renewable resource is a _____ of resource which can only be used once and cannot be replaced.

11. A _____ is a power tool consisting of a pointed copper head attached to an iron bar. It is heated electrically or in a blow lamp, etc., and used to melt the solder in soldering small parts together.
12. A _____ is a metal device used for holding parts together. It has a threaded body and is often tapered. It is rotated by means of a slot or cross in its head. The tool used to turn one is called a _____. A _____ has a cross-shaped point.
13. An electron _____ is an instrument which enables scientists to see objects invisible to the naked eye.
14. A _____ (AE) or _____ (BE) is a device used for tightening or loosening a nut.
15. Flash welding is a _____ of electric welding in which a high voltage is applied to two metal parts to be joined so that an arc forms between their surfaces. The current is then turned off and pressure is applied to the parts to finish the welding.
16. A _____ (AE, *torch* in BE) is a portable electric light, powered by a battery. It is usually cylindrical in shape.
17. "Top dead-center" is the _____ of a piston in an internal-combustion engine when it has reached the top of its stroke.
18. A _____ (AE, *windscreen* in BE) is a sheet of glass that forms the front window of a motor vehicle.
19. The wheelbase is the _____ between the front and back axles of a motor vehicle.
20. A Fresnel lens is a _____ with one face cut into a series of steps. It enables a relatively light and robust lens of short focal length though poor optical quality to be used in projectors, searchlights, spotlights, car headlights, and lighthouses.

This video might be of interest:

<https://www.youtube.com/watch?v=A4OmtyaBHFE>

To see Germany's future, look at its cars | The Economist, 11 min, 2021.